DATA ANALYTICS PORTFOLIO PROJECT · NOVEMBER 2026

AdventureWorks Sales Analytics

A five-page Power BI solution analyzing 60,398 sales transactions to identify revenue drivers, profitability trends, customer behavior, and territory performance — while maintaining full reconciliation to source data.

$307.09M	$126.29M	41.12%	27,659	18,484
REVENUE	GROSS PROFIT	GROSS MARGIN	ORDERS	CUSTOMERS

> Identified the products, territories, and customers driving a $307M sales business while maintaining 100% reconciliation to source data.

REPORT CONTENTS

Business problem · data model and architecture · five dashboard pages · key insights · business impact · technical skills · validation and reconciliation · next steps.

TOOLS USED

Power BI Desktop · DAX · Power Query · Azure SQL Database (reference integration) · T-SQL example · Data Modeling

Source AdventureWorksDW.xlsx — 6 tables, 60,398 fact rows
Model 7 tables, 6 relationships, 65 DAX measures

Report AdventureWorks Sales.pbix, current Desktop version
Five visible pages · 60,398 order lines · November 2026
Azure SQL is documented as a deployment path, not a live PBIX connection.

The business problem: a wholesaler that can price nothing and see everything

AdventureWorks sells bicycles, components and accessories to resellers in six countries. Every order is already recorded, yet the commercial team cannot answer the questions that decide next quarter: which product lines actually earn their margin, which territories are carrying the business, and whether 18,484 customers are a durable base or a one-off spike.

Why the raw workbook cannot answer them

The source is a single Excel workbook with one 60,398-row fact sheet and five dimensions. It is accurate and complete, but it is a record, not an instrument: no relationships, no measures, no calendar the facts agree with, and amounts stored **per unit** so that a plain `SUM` of the sales column silently understates revenue.

There is also no product hierarchy, no promotion or currency dimension, and no returns, budget or quota data. The report therefore has to be explicit about what it can prove and what it cannot — and it has to reconcile, to the cent, with the workbook it came from.

Where the margin is

Rank product lines and products on profit, not revenue, so the 2% line with the best margin is visible.

Where the demand is

Compare territory groups and countries on the same basis, without mixing two filter paths.

Who the buyer is

Separate one-off buyers from repeat accounts and size the value of each customer relationship.

Questions the report answers

01 What is revenue, gross profit and margin — and what is the order value behind them?

02 Which product line and which individual products produce the profit?

03 How concentrated is revenue across countries and territory groups?

04 How many buyers return, and what is a customer worth over their lifetime?

05 Does fulfilment meet its lead-time commitment, line by line?

06 Can any single figure be traced back to the exact order lines that produced it?

DESIGN RULE

Reconcile, don't approximate. Every page carries the calculation rule in its footer — revenue is `SUMX(UnitPrice × OrderQuantity)` — and the Detail page totals to the same $307,092,945.20 the KPI cards show.

DATA MODEL & ARCHITECTURE · 1 OF 3

Source, grain and data quality

AdventureWorksDW.xlsx — a read-only extract of the AdventureWorks internet-sales schema. All figures in this document were recomputed from it and cross-checked against the model and the browser dashboard.

SHEET	ROWS	COLS	ROLE AND KEY
FactInternetSales (table `Table2`)	60,398	26	Grain: order number + line number (unique). Every key resolves to a dimension.
DimProduct	606	36	`ProductKey` 1–606; only 158 products are ever sold.
DimCustomer	18,484	29	`CustomerKey`; `GeographyKey` → DimGeography.
DimGeography	655	11	`GeographyKey`; 336 are used by customers.
DimSalesTerritory	11	6	`SalesTerritoryKey` 1–11; the fact uses 1–10.
DimDate	2,191	19	Daily calendar 2005-01-01 → 2010-12-31; `DateKey` = YYYYMMDD.

TRAP 1 · AMOUNTS ARE PER UNIT

`SalesAmount`, `ExtendedAmount` and `TotalProductCost` equal the *unit* price and unit cost on every row; `OrderQuantity` (1–20) is not baked in. Tax is 8% and freight 2.5% of the unit amount. A plain `SUM(SalesAmount)` would understate revenue by roughly an order of magnitude, so every money measure multiplies by quantity.

TRAP 2 · TWO DATE TIMELINES

The date *keys* (2005-07 → 2008-07) match DimDate. The date *columns* are re-based to 2021-01-01 → 2022-12-31 — 2021 holds the 2005+2006 rows (3,690 lines) and 2022 holds 2007+2008 (56,708). The report uses the 2021–2022 display timeline with its own calendar and never relates the shifted columns to DimDate.

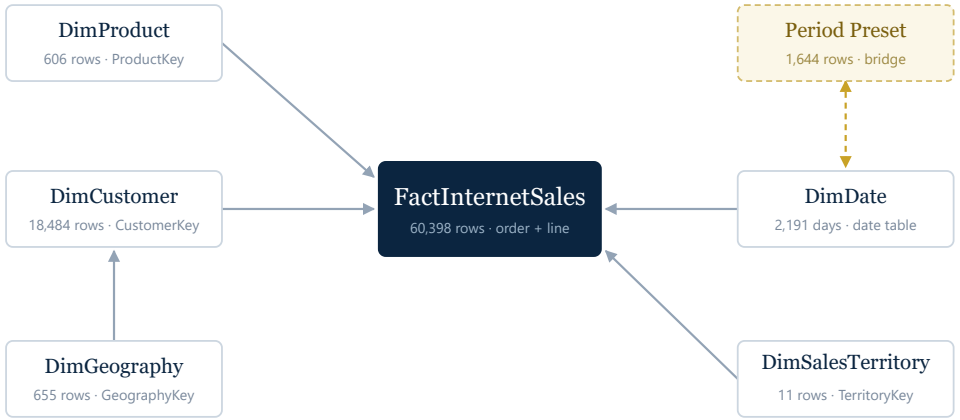
What the workbook does not contain. No product subcategory or category sheet (`ProductSubcategoryKey` 1–37 has nothing to join to), no promotion or currency dimension (so no currency conversion is applied), and no returns, refunds, budget or quota data. Those keys are dropped rather than given invented relationships, and no figure in this report is modelled from them.

DATA MODEL & ARCHITECTURE · 2 OF 3

Relationships, calendar and filter architecture

Seven tables, six relationships, one marked date table, 65 measures in eight display folders. The model is embedded in the .pbix, so the report opens with no refresh prompt.

STAR SCHEMA — ONE FACT, FIVE DIMENSIONS



Six relationships. Four run dimension → fact on the matching key — Date → OrderDate, plus ProductKey, CustomerKey and SalesTerritoryKey — so a slicer narrows the fact and never the reverse. A fifth snowflakes DimGeography into DimCustomer; the sixth is the Period Preset bridge, bidirectional only because DimDate sits on its many side. Territory never joins through Customer → Geography, so it cannot be filtered two ways.

EXTRACT AND LOAD

The workbook is never written back to. A Python step reads it with stdlib XML streaming and writes seven curated CSVs (fact plus five dimensions and the preset bridge); each table is loaded through an M `File.Contents` partition, so the packaged report is self-contained. Zero-length strings are normalised to null, all-null columns (`CarrierTrackingNumber`, `CustomerPONumber`, `LargePhoto`, `SalesTerritoryImage`) and the eight duplicate-language description columns are dropped.

DATE DECISION

The 2021–2022 display timeline is the reporting calendar. DimDate is rebuilt to run 2021-01-01 → 2022-12-31 and is marked as the date table; the raw YYYYMMDD keys are hidden. This keeps every visual, the Period slicer and the time-intelligence measures on one timeline. Customer first-purchase and product start/end dates stay unusable for time intelligence because they remain on the source timeline — they are referenced but never used to slice.

FILTER BAR — WEB PARITY

All five pages open with the same control strip above the KPI row. It is real, single-source filtering: the period range and the preset intersect (AND), and the reset button is Desktop's native *Clear all slicers*, so no bookmark state can drift.

Period — date range · Preset — All / 2021 / 2022 / H2 2022 · Country · Product line · Region · Gender · Income band · Lead-time band (Detail) · Reset filters

WHY A BRIDGE TABLE

Calculation groups re-evaluate measures but never the rows a chart draws, so the presets are a 1,644-row (preset, date) bridge. Its relationship to DimDate must stay bidirectional: filter direction runs from the one side to the many side, and here the calendar is the many side — one-directional, picking *2021* changed nothing.

DATA MODEL & ARCHITECTURE · 3 OF 3

DAX measures

Measures live on the fact table and are grouped by intent, not by visual. The amount rule — unit price and unit cost multiplied by quantity — is applied once, at the base.

01 Core 8 | 02 Customers 5 | 03 Fulfillment 4 | 04 Time intelligence 5 | 05 Context 4 | 06 Geography 3 | 07 KPI sub-lines 20 | 08 KPI values 16

REVENUE 01 Core

```
Revenue =
SUMX ( FactInternetSales,
       FactInternetSales[UnitPrice]
         * FactInternetSales[OrderQuantity] )
```

GROSS PROFIT & MARGIN 01 Core

```
Gross Profit =
SUMX ( FactInternetSales,
       ( FactInternetSales[UnitPrice]
         - FactInternetSales[UnitCost] )
           * FactInternetSales[OrderQuantity] )

Gross Margin % =
DIVIDE ( [Gross Profit], [Revenue] )
```

REPEAT RATE 02 Customers

```
Repeat Customers =
COUNTROWS (
    FILTER (
        SUMMARIZE ( FactInternetSales,
            FactInternetSales[CustomerKey],
            "OrderCount",
            DISTINCTCOUNT (
                FactInternetSales[SalesOrderNumber] ) ),
        [OrderCount] >= 2 ) )

Repeat Rate =
DIVIDE ( [Repeat Customers], [Customers] )
```

LATE LINE RATE — ZERO-SAFE 03 Fulfillment

```
Late Lines =
CALCULATE ( COUNTROWS ( FactInternetSales ),
    KEEPFILTERS ( FactInternetSales[LateDays] > 0 ) ) + 0
Late Line Rate = DIVIDE ( [Late Lines], [Lines], 0 )
```

KPI REVENUE — ONE FORMATTING PATTERN FOR EVERY MONEY CARD 08 KPI values

```
// a text measure: the card's display units turned 27,659 into "28K"
KPI Revenue = VAR v = [Revenue]  VAR a = ABS ( v )
RETURN "$" & SWITCH ( TRUE (),
    a >= 1000000000, FORMAT ( v / 1000000000, "0.00" ) & "B",
    a >= 1000000,    FORMAT ( v / 1000000,    "0.00" ) & "M",
    a >= 1000,       FORMAT ( v / 1000,       "0.0" )  & "K",
    FORMAT ( ROUND ( v, 0 ), "#,##0" ) )
```

Three deliberate choices

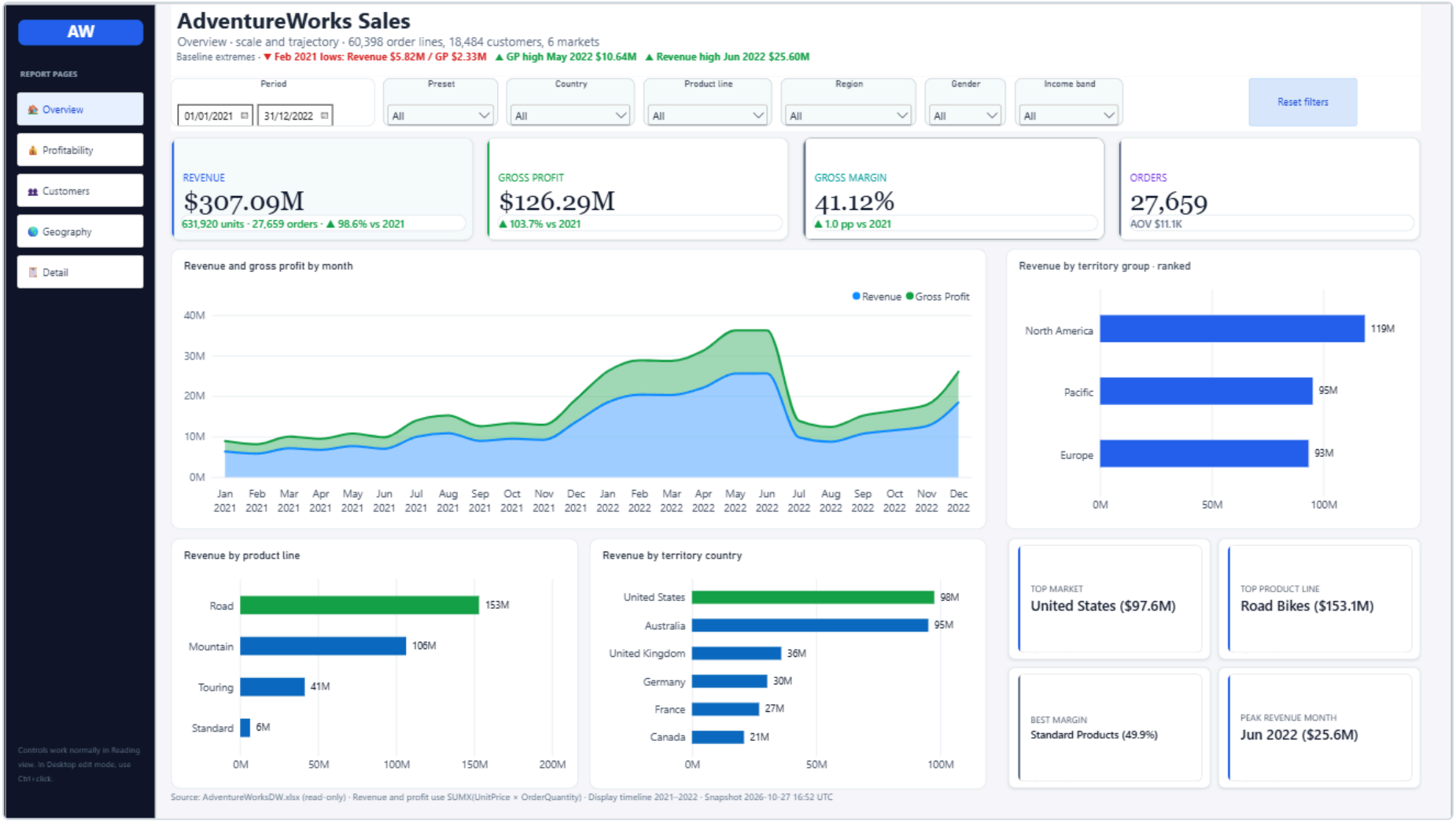
1. SUMX, never SUM. Quantity is part of the amount, so it is applied inside the iterator; cost and freight follow the same rule, which is why margin is 41.12%.

2. +0 on empty results. Every LateDays value is zero, so the card shows **0.00%** and **0** — the truth, not a blank that looks like a formatting error.

POWER BI DESKTOP · 01 OF 05

Overview

Revenue, profit and margin KPIs with year-over-year indicators, monthly trend, territory and country rankings, product mix, and market/product insights.



HEADLINE ROW

$307.09M revenue, $126.29M gross profit and 41.12% margin; revenue grew 98.6%, profit grew 103.7%, and margin improved 1.0 pp vs 2021.

BUSINESS MIX

North America leads territory groups at $119M; Road leads product-line revenue at about $153M. United States is the top country market.

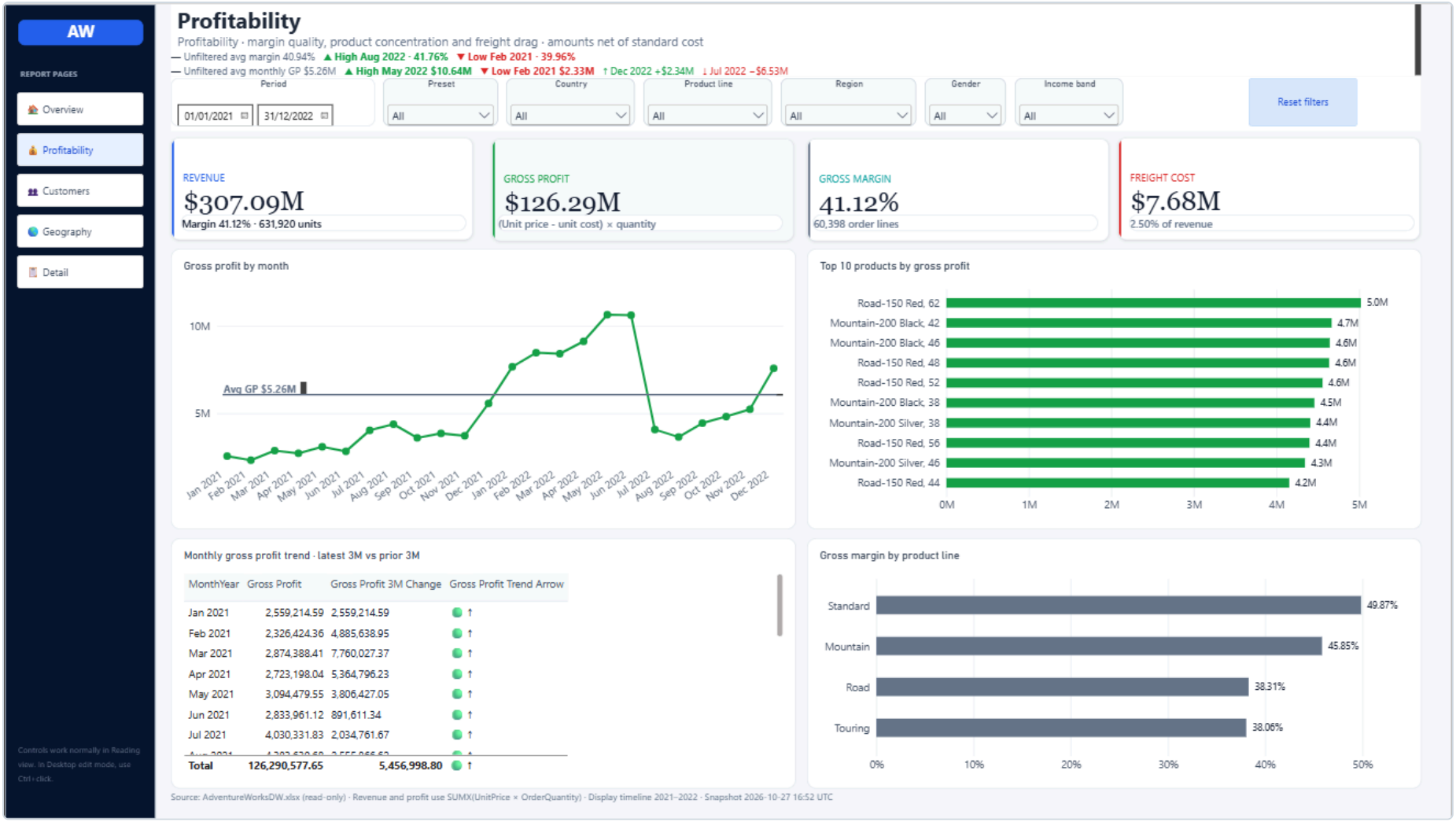
CONTROLS

Shared period, preset, country, product, territory and customer filters, with a reset control and dynamic KPI and chart updates.

POWER BI DESKTOP · 02 OF 05

Profitability

Monthly gross profit, top products by gross profit, a recent-three-month trend table, and gross margin by product line.



MONTHLY TREND

Gross profit is compared with a $5.26M blended monthly reference; the 2022 peak is followed by a mid-year decline and a year-end recovery.

PROFIT, NOT REVENUE

The top-10 list ranks products by gross profit. Standard is the highest-margin product line at about 49.9%.

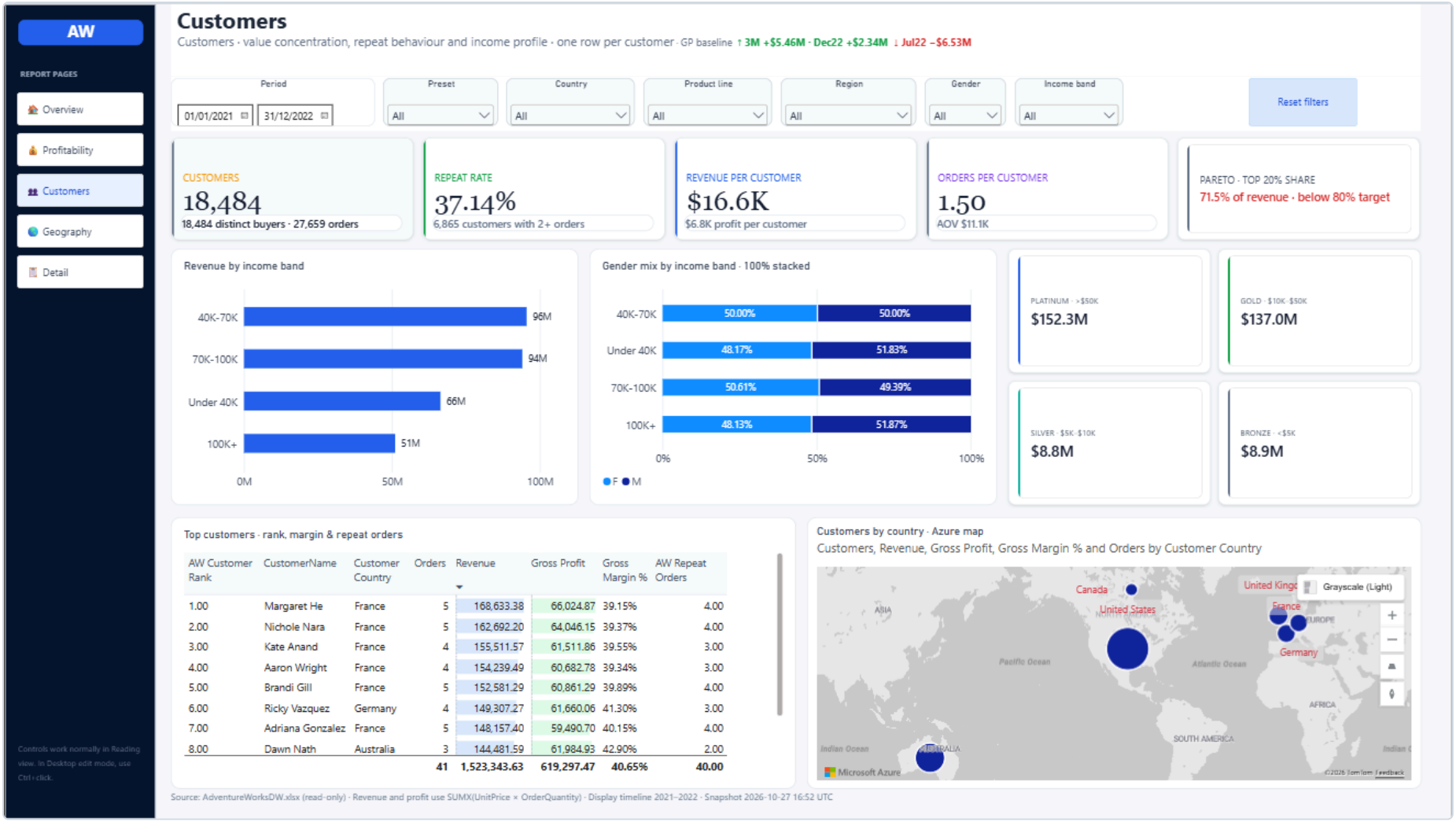
COST CONTEXT

Revenue, gross profit, margin and freight are shown together; freight totals $7.68M (2.50% of revenue).

POWER BI DESKTOP · 03 OF 05

Customers

Customer repeat behavior, income and gender mix, top-customer detail, customer-country map, and the share of revenue from the top 20% of customers.



REPEAT BEHAVIOR

6,865 of 18,484 customers placed two or more orders, a 37.14% repeat rate across 27,659 orders.

PARETO CHECK

The top 20% of customers contribute 71.5% of revenue, below the 80% benchmark shown on the page.

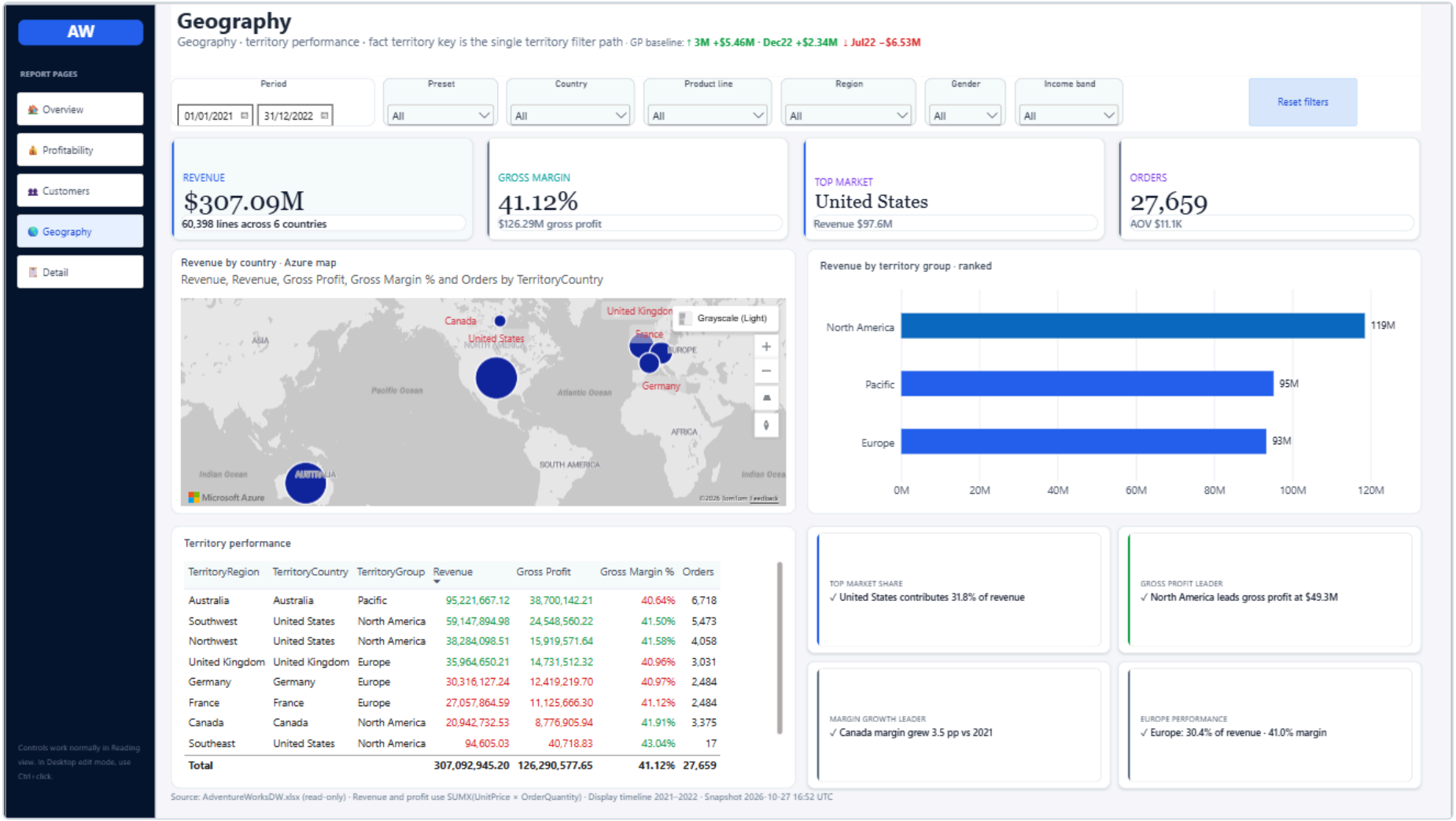
CUSTOMER VALUE

Revenue averages $16.6K per customer; the customer table and map support ranking and geographic exploration.

POWER BI DESKTOP · 04 OF 05

Geography

Revenue by country and territory group, an Azure Maps visual, territory-level performance detail, and geography insights.



TERRITORY MIX

North America is largest at about $119M, followed by Pacific ($95M) and Europe ($93M).

PERFORMANCE TABLE

Revenue, gross profit and gross margin use green/red formatting to distinguish above- and below-benchmark territory results.

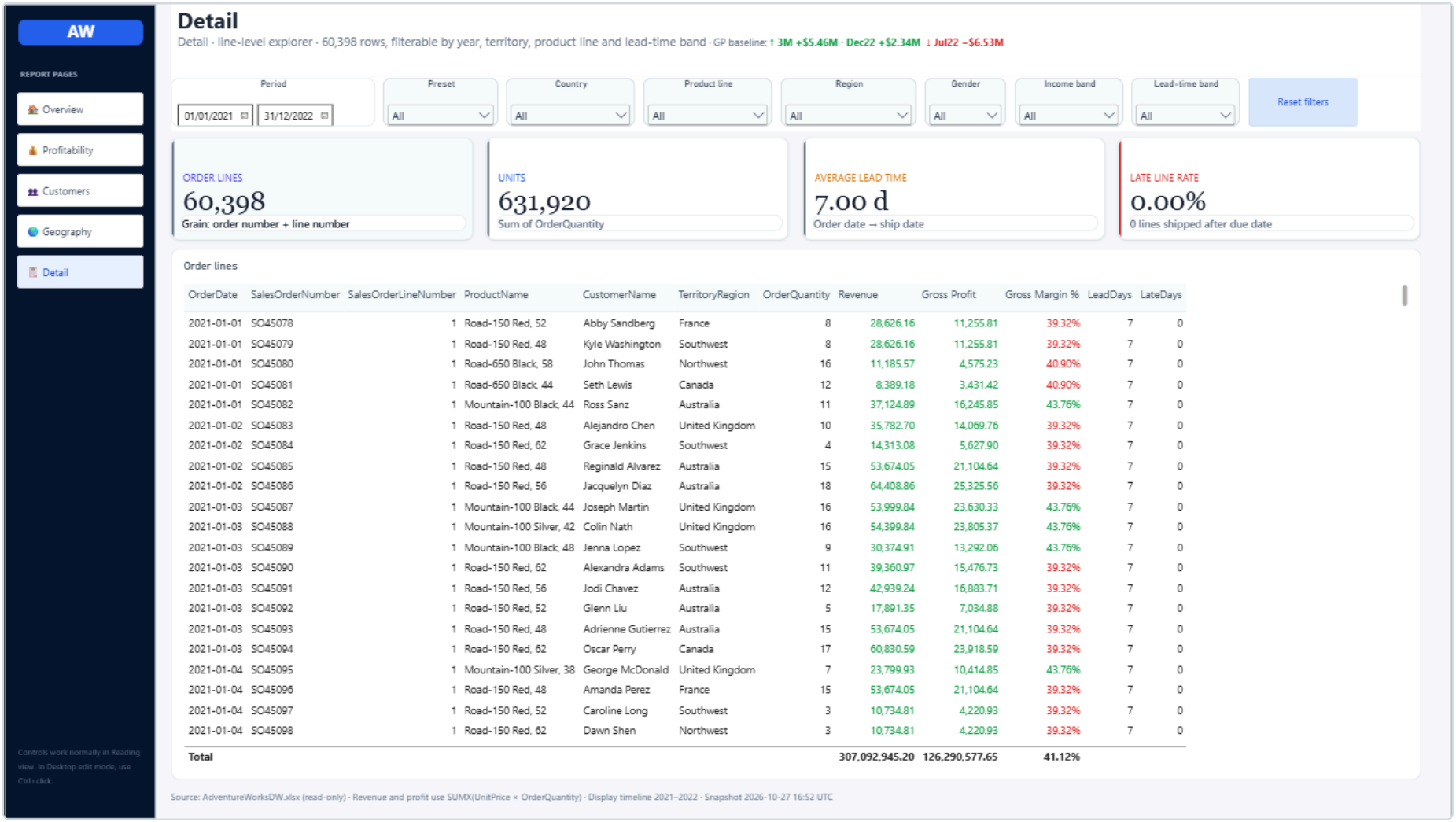
MAP AND FILTERS

Azure Maps and report slicers support geographic exploration while keeping the fact-territory and customer-country paths separate.

POWER BI DESKTOP · 05 OF 05

Detail

A filterable, line-level order explorer with revenue, gross profit, margin and fulfillment fields, plus totals that reconcile to the KPI cards.



OrderDate	SalesOrderNumber	SalesOrderLineNumber	ProductName	CustomerName	TerritoryRegion	OrderQuantity	Revenue	Gross Profit	Gross Margin %	LeadDays	LateDays
2021-01-01	SO45078	1	Road-150 Red, 52	Abby Sandberg	France	8	28,626.16	11,255.81	39.32%	7	0
2021-01-01	SO45079	1	Road-150 Red, 48	Kyle Washington	Southwest	8	28,626.16	11,255.81	39.32%	7	0
2021-01-01	SO45080	1	Road-650 Black, 58	John Thomas	Northwest	16	11,185.57	4,575.23	40.90%	7	0
2021-01-01	SO45081	1	Road-650 Black, 44	Seth Lewis	Canada	12	8,389.18	3,431.42	40.90%	7	0
2021-01-01	SO45082	1	Mountain-100 Black, 44	Ross Sanz	Australia	11	37,124.89	16,245.85	43.76%	7	0
2021-01-02	SO45083	1	Road-150 Red, 48	Alejandro Chen	United Kingdom	10	35,782.70	14,069.76	39.32%	7	0
2021-01-02	SO45084	1	Road-150 Red, 62	Grace Jenkins	Southwest	4	14,313.08	5,627.90	39.32%	7	0
2021-01-02	SO45085	1	Road-150 Red, 48	Reginald Alvarez	Australia	15	53,674.05	21,104.64	39.32%	7	0
2021-01-02	SO45086	1	Road-150 Red, 56	Jacquelyn Diaz	Australia	18	64,408.86	25,325.56	39.32%	7	0
2021-01-03	SO45087	1	Mountain-100 Black, 44	Joseph Martin	United Kingdom	16	53,999.84	23,630.33	43.76%	7	0
2021-01-03	SO45088	1	Mountain-100 Silver, 42	Colin Nath	United Kingdom	16	54,399.84	23,805.37	43.76%	7	0
2021-01-03	SO45089	1	Mountain-100 Black, 48	Jenna Lopez	Southwest	9	30,374.91	13,292.06	43.76%	7	0
2021-01-03	SO45090	1	Road-150 Red, 62	Alexandra Adams	Southwest	11	39,360.97	15,476.73	39.32%	7	0
2021-01-03	SO45091	1	Road-150 Red, 56	Jodi Chavez	Australia	12	42,939.24	16,883.71	39.32%	7	0
2021-01-03	SO45092	1	Road-150 Red, 52	Glenn Liu	Australia	5	17,891.35	7,034.88	39.32%	7	0
2021-01-03	SO45093	1	Road-150 Red, 48	Adrienne Gutierrez	Australia	15	53,674.05	21,104.64	39.32%	7	0
2021-01-03	SO45094	1	Road-150 Red, 62	Oscar Perry	Canada	17	60,830.59	23,918.59	39.32%	7	0
2021-01-04	SO45095	1	Mountain-100 Silver, 38	George McDonald	United Kingdom	7	23,799.93	10,414.85	43.76%	7	0
2021-01-04	SO45096	1	Road-150 Red, 48	Amanda Perez	France	15	53,674.05	21,104.64	39.32%	7	0
2021-01-04	SO45097	1	Road-150 Red, 52	Caroline Long	Southwest	3	10,734.81	4,220.93	39.32%	7	0
2021-01-04	SO45098	1	Road-150 Red, 62	Dawn Shen	Northwest	3	10,734.81	4,220.93	39.32%	7	0
Total							**307,092,945.20**	**126,290,577.65**	**41.12%**		

GRAIN AND COLOR

Order number plus line number is unique across 60,398 rows. Revenue and gross profit are green/red against selected-row averages; margin is colored against the 41.12% baseline.

RECONCILIATION

The Total row reads **$307,092,945.20** revenue and **$126,290,577.65** gross profit — matching the KPI cards to the cent.

FULFILLMENT

Units total 631,920, average lead time is 7.00 days, and late-line rate is 0.00%; the lead-time band filter is specific to this page.

FULL SELECTION · ALL 60,398 LINES

Key insights

Every figure below is a control total recomputed from the extract, the model and the browser rendering — all three agree to the cent.

41.12%

BLENDED GROSS MARGIN

$126.29M profit on $307.09M revenue, held across 631,920 units at $485.97 each.

37.14%

REPEAT-CUSTOMER RATE

6,865 of 18,484 buyers placed two or more orders — a reseller base, not a one-off spike.

$11,102.82

AVERAGE ORDER VALUE

27,659 orders, 1.50 per customer: few, large, repeatable transactions.

49.87%

STANDARD-LINE MARGIN

The smallest line by revenue ($6.33M, 2.06%) is the most profitable one.

One line pays for half the business.

Road contributes $153.08M (49.85% of revenue); Mountain adds $106.32M (34.62%) and Touring $41.36M (13.47%). Three lines carry 98% of revenue, so assortment risk is concentrated by design.

Three territory groups, no dependency.

North America $118.53M (38.60%), Pacific $95.22M (31.01%), Europe $93.34M (30.39%) — an 8.2-point spread. But two countries, the United States ($98.61M) and Australia ($91.93M), are 62% of revenue.

The 2021–2022 jump is a calendar, not growth.

$102.85M over 3,690 lines in 2021 against $204.24M over 56,708 in 2022. The display timeline folds four source years into two, so YoY is shown for completeness but never read as like-for-like demand.

Fulfilment is a policy, not a risk.

Average lead time is 7.00 days and 0 of 60,398 lines shipped after their due date. The late-line card is a control that proves the process held — it cannot be used to argue service improved.

Cost structure is fixed by the source.

Freight is exactly 2.50% of revenue ($7.68M) and tax 8% of the unit amount. Neither is a lever the business can pull, so the report reads margin on unit price and unit cost only.

Value per customer is stable, not spiky.

$16.6K revenue and $6.8K profit per customer across 18,484 accounts. The top 20% of customers contribute 71.5% of revenue, below the 80% reference; the report makes that concentration visible.

READ WITH CARE

No returns, refunds, budget or quota data exists in the workbook, so every figure here is shipped sales. There is no target variance, no revenue recognised net of returns, and no currency conversion.

WHAT THE REPORT CHANGES

Business impact

The figures are the evidence; the decisions are the point. Each impact below rests on a number that could not be produced from the workbook as it stood.

Margin is now a product decision

Road is half the revenue ($153.08M, 49.85%), yet Standard — 2.06% of revenue — returns the best margin at 49.87%. A revenue-only view keeps funding the biggest line; this one shows which line earns.

Concentration is measured

Three territory groups land within 8.2 points (38.60% / 31.01% / 30.39%), but the United States and Australia are 62% of revenue between them. Territory investment can now be argued with a number.

Retention has a baseline

6,865 of 18,484 buyers return (37.14%) at $16.6K revenue and $6.8K profit per account. Any account programme can be measured against that line instead of intuition.

Service is provable, not anecdotal

7.00-day average lead time and 0 late lines across 60,398 rows. Fulfilment now has a control that holds at line level and would move the moment the process slipped.

THE QUESTION	IN THE WORKBOOK	IN THE REPORT
What did we earn?	A sum of a per-unit column, understated by the quantity factor	`SUMX(UnitPrice × Quantity)` — $307,092,945.20 on every page
Which line earns?	No margin field and no hierarchy to rank	Gross profit and margin by line and product; best line 49.87%
Where is the demand?	Two territory paths that disagree, six countries unranked	One agreed path per visual: 3 groups, 6 countries, 10 territories
Who is the buyer?	18,484 customer rows with no repeat or value logic	37.14% repeat rate; top 20% contribute 71.5% of revenue; $16.6K revenue per customer
Does service hold?	No lead-time field and no late-delivery flag	7.00-day average lead time, 0 late lines of 60,398
Can we trust it?	Manual cross-checks, no reconciliation trail	10/10 integrity checks; the Detail footer reconciles to the cent

THE NET EFFECT

A spreadsheet that could not state revenue correctly has become a model that states it once, provably, on every page — and that a commercial reader can filter, drill and challenge without opening Excel.

One workbook in, one model out 5 pages · 65 measures · 7 tables · 6 relationships · 60,398 lines, none dropped · every figure traced back to the row that produced it.

HOW IT WAS BUILT

Technical skills demonstrated

Everything below was done from a single read-only workbook, with no database, no external service and no manual refresh between captures.

Data modelling & preparation

- **Star schema** — one fact and five dimensions, six relationships with explicit cardinality, and a grain (order + line) proven unique across 60,398 rows.
- **Date-table discipline** — a dedicated 2021–2022 calendar marked as the date table, with raw YYYYMMDD keys hidden so no visual can slice on the wrong field.
- **Filter-direction design** — diagnosing why a single-direction relationship silently disabled the period presets, and keeping the bridge bidirectional on purpose.
- **Power Query (M)** — `File.Contents` partitions, type enforcement, zero-length strings normalised to null, all-null and duplicate-language columns pruned.
- **Source discipline** — extract to curated CSVs, never write back, and drop keys that have no dimension rather than inventing a join.
- **Extraction pipeline** — Python parses the workbook with stdlib zip/XML streaming and emits seven curated CSVs: one per table plus the preset bridge.

DAX & analytical logic

- **Iterator-based aggregation** — `SUMX` over unit price and unit cost, because quantity is not baked into the amount columns.
- **Safe ratios** — `DIVIDE` with an alternate result everywhere a denominator can be empty; margin, AOV and rates never divide by zero.
- **Zero-safe counting** — `KEEPFILTERS` plus `+ 0` so a genuinely empty result reads 0, not blank.
- **Time intelligence** — `SAMEPERIODLASTYEAR` for prior-year revenue, profit and margin, with the change expressed in percentage points.
- **Context and formatting** — `DISTINCTCOUNT`, `SUMMARIZE`, `FILTER`, `SELECTEDVALUE`, and `SWITCH`-based text measures that control exactly how a card reads.
- **Measure architecture** — 65 measures in eight display folders, where text KPIs reuse the base measures so a formatting change can never fork the logic.

Report, engineering & QA

- **Report design** — five-page PBIR layout, a shared filter bar on every page, layered KPI cards, and a provenance footer that states source, rule, timeline and snapshot.
- **Interaction design** — slicer actions (`ClearAllSlicers`), page navigation, cross-filtering, and a Detail explorer that reconciles to the summary cards.
- **Code-generated assets** — Python generators for the TMDL model and PBIR pages, plus a field-reference check that validates every visual against the model.
- **Reconciliation testing** — control totals recomputed from source, model and rendering, 10/10 integrity checks, and the amount rule re-tested on every rebuild.
- **Evidence capture** — full-resolution captures of the current Power BI Desktop report pages, cropped to the report canvas.
- **Documentation** — a README recording the page map, control totals, rebuild steps and every known limitation, so another analyst can audit or rebuild the model.

TOOLKIT Power BI Desktop · Power Query (M) · DAX · PBIP (TMDL + PBIR) · Python 3.13 (stdlib XML/zip parsing) · Excel · Azure SQL Database/T-SQL reference integration · Edge for PDF authoring.

What that adds up to a model that cannot silently misstate revenue, a report a non-technical reader can operate unaided, and a documented trail from the source row to the number on the screen.

CONTROLS, TIE-OUT AND CAVEATS

Validation & reconciliation

Ten integrity checks run on every open — rows, revenue, profit, units, orders, customers, line uniqueness, the amount rule, key resolution and late lines. Current state: 10/10 pass, and every KPI ties back to the order lines that produced it.

CONTROL TOTALS — VERIFIED

Order lines	**60,398**
Units	**631,920**
Revenue	**$307,092,945.20**
Gross profit	**$126,290,577.65**
Gross margin	**41.12%**
Orders	**27,659**
Customers	**18,484**
Repeat customers	**6,865 (37.14%)**
Average order value	**$11,102.82**
Freight cost	**$7,677,351.65 (2.50%)**
Average lead time	**7.00 days**
Late lines	**0**
Countries / territories / groups	**6 / 10 / 3**
2021 revenue	**$102,854,715.37** (3,690 lines)
2022 revenue	**$204,238,229.83** (56,708 lines)

LIMITATIONS — VERIFIED AGAINST THE WORKBOOK

- **Per-unit amounts.** Any new measure must multiply by `OrderQuantity`; `SUM(SalesAmount)` is never used.
- **Two timelines.** Customer first-purchase and product start/end dates stay on the source calendar and are not used for time intelligence.
- **No product hierarchy.** Subcategory and category names are not recoverable from this workbook, so product analysis stops at the line and product.
- **No promotion or currency dimension.** Keys are dropped; all amounts are in one reporting currency with no conversion applied.
- **No returns, budget or quota.** Shipped sales only — no target variance.
- **Territory is reachable twice.** The fact key drives territory visuals and the customer dimension drives country visuals; mixing them in one visual would create an ambiguous path.
- **Dimension coverage.** Only 158 of 606 products sell, and territory 11 exists but is never used, so counting the catalogue requires an outer join.

THE TIE-OUT

$307,092,945.20 on the Overview card, $307,092,945.20 in the Detail footer, and the same figure when the extract is recomputed from the workbook — three independent paths to one number. That is the standard every figure in this report is held to.

WHERE THIS GOES NEXT

Next steps

Four additions that would turn a finished analytical model into a governed, decision-grade product. Each is additive — none requires reopening what has already been reconciled.

Restore the hierarchy

Source subcategory and category names externally, then add a two-level product drill that keeps the 158-product filter intact.

Unlocks category-level margin — the one dimension the workbook cannot supply.

Add targets

Load budget or quota by territory and month to convert margin reporting into variance reporting.

Turns "how did we do?" into "did we hit the plan, and where did we miss?"

Implement Row-Level Security

Apply row-level security on territory group and publish the report as an app with a scheduled refresh.

Lets a sales director see their region without exposing the other two.

Deepen the drill

Promote the Detail page to true drillthrough from the product and country visuals, keeping the reconciling footer.

Moves a reader from "that number looks wrong" to the exact lines in one click.

WHAT SHIPS WITH THIS PROJECT

AdventureWorks Sales.pbix

The packaged report: five visible pages, model embedded, opens with no refresh prompt.

pbi/ — editable source

TMDL semantic model and PBIR report definition, rebuildable from the workbook.

dashboard/ — reference build

Dependency-free HTML dashboard carrying the ten integrity checks used to test the model.

portfolio/ — this document

HTML source, five current Power BI Desktop captures, and the rendered PDF.

01

Keep the grain

Every new page stays at one row per order line — `SalesOrderNumber` plus `SalesOrderLineNumber`. Because revenue is a `SUMX` over quantity, a table that fans the fact out would inflate $307.09M without raising a single error.

02

Isolate the targets

A budget table must reach the fact through the territory and calendar dimensions, never directly. Joined at line level the quota is multiplied by the number of orders, and every variance figure becomes confidently wrong.

03

Reuse the path

Territory is reachable from the fact and again via customer → geography. Pick one per visual. Enabling both reintroduces the ambiguous filter path this model was built to remove.

The model is finished and reconciled. What it needs next is richer context — hierarchy, targets, security — not repair.